

Predicting Canadian Residential Housing Prices Using Multiple Linear Regression and Random Forest Models

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Introduction

The residential real estate in Canada's market has gone through a lot of changes over the last ten years. Demographics transformations such as population growth, urbanization, waves of immigration, shifting interest rates, and changing buyer preferences have all played a role. All these points together, made significantly harder for homebuyers, investors, developers, and policymakers to get a clear picture of what is actually driving property values and housing demand. Real estate remains one of the largest asset classes in the country, which means the stakes around how decisions get made are genuinely high.

That's where data science and business analytics have started to shift things. It is now realistic to work through real estate data at a scale that would've been overwhelming or just plain impractical not too long ago. Working with detailed property-level data such as square footage, property type, bedroom and bathroom counts, nearby amenities, location, patterns begin to emerge that aren't obvious otherwise. Consequently, it is possible to identify that what's actually driving price. For anyone trying to get a honest read on market conditions or pressure-test a valuation, that kind of grounded analysis makes a real difference.

This study takes that approach and applies it to a large dataset of Canadian residential properties. The goal is to dig into what is influencing housing prices and property valuation, and to show how working through that data analytically can surface real business intelligence, the kind that makes a tangible difference in how real estate decisions get made.

Business Context

Few sectors touch the Canadian economy quite like real estate, since it drives employment, fuels investment, and for millions of households, representing the single largest piece of their financial picture. But over the past several years, affordability has quietly shifted from a niche

policy concern into something much more pressing, especially in cities that have grown faster than their housing stock could keep up with. And it is not a uniform story either; what's happening in Vancouver or Toronto looks very different from what find in smaller provinces, where local economies, demographic trends, and the pace of new construction have all pushed property values in their own directions.

One thing that has changed across the board, though, is how much data is now available and what individuals are doing with it. Real estate firms, banks, investment groups, and government bodies have all started leaning on analytics in ways they did not work with decade ago. Additionally, predictive models are being used to estimate significant analyses and outcomes such as property values, flag investment opportunities, and even inform housing policy.

The dataset behind this project draws residential listings from across multiple Canadian provinces and territories. It goes beyond the basics such as standard property details, it brings in geographic context and amenity-related variables that together give a well-rounded view of the housing landscape. All of that combined makes it a solid starting point for digging into what's actually driving residential prices and where the most defined market patterns tend to show up.

Research Problem

Pricing a home is not as simple as tallying up bedrooms and checking the postal code. There are layers to it, including location, size, amenities, and property type; while most buyers and investors have a general sense that these things matter, pinning down how much each one matters is a lot harder. The relationship between these variables shifts depending on where you are in Canada and what kind of property it is being analysed, which makes it difficult to reason about price without actually working through the data.

That's essentially the gap this study is trying to fill. There's no shortage of opinions about what drives Canadian housing prices, but evidence-based, data-driven analysis of the specific factors at play and how it rank in terms of influence is harder to come by. By working through a large housing dataset, this project takes a closer look at what's actually shaping property values and aims to produce insights that are grounded in something more solid than intuition or anecdote.

Background Research & Literature Review

House Price Prediction

House price prediction in Canada has become important in real estate analytics due to housing market volatility (high percentage of temporary residents, government regulations, and economic factors). Over the past 9 years, Canada has experienced an increase in housing sales, mainly from 2018 to 2021, when sales reached their maximum number of units sold by year. This event could be explained by a combination of economic, social and monetary policy factors. During the pandemic, Bank of Canada reduced interest rates to support the economy, consequently, buyers borrowed money as the mortgage rates became low. After the pandemic, many buyers who were afraid of future uncertainty began purchasing again. This also affected house prices as demand was high at that time. The average house price reached a peak in 2022, and returned to increase in 2025, especially in British Columbia and Ontario, where the average house price was higher than the average price in Canada. Many buyers took advantage of this to invest in a house to rent, and use the rental income to pay back the mortgage.

Overall, the house price prediction is important for understanding which factors drive house prices and the price difference between cities and provinces in Canada.

House price forecast – MLR

Multiple Linear Regression is a statistical method used to predict a dependent variable (outcome) and two or more variables (factors that affect the outcome). For our analysis, the dependent variable will be the house price and the independent variables will be the other factors. The model assumes a linear relationship between predictor and independent variables.

Studies emphasized the relevance of Multiple Linear Regression for house price forecasting due to its interpretability and simplicity, as well as being the most suitable regression model for structured data with linear relationships.

House price forecast – Random Forest

Random Forest is a machine learning algorithm, that combines the output of multiple decision trees to achieve a single result (IBM, n.d.). Instead of assuming just a simple trend, Random Forest uses an ensemble of decision trees to forecast and estimate future values. It is used in housing price prediction as it captures non-linear relationships between variables. Čeh et al. (2018) used a Random Forest model to predict apartment prices and found that it performed better than traditional multiple regression in terms of accuracy.

Models Comparison

In this case, MLR and Random Forest are excellent techniques that can be used for predicting house prices in Canada. However, they differ significantly, as we can see in the table below:

Table 1 – MLR and Random Forest

Type	MLR	Random Forest
Complexity	Low	High
Interpretation	High	Low
Non-linear relationships	Low	High
Accuracy	Moderate	High
Flexibility	Low	High

MLR is less complex and easier to interpret than Random Forest, as MLR struggles with outliers, non-linear distribution and missing patterns. On the other hand, Random Forest is more accurate at predicting values as it takes into consideration the interactions between variables than MLR.

Industry Report

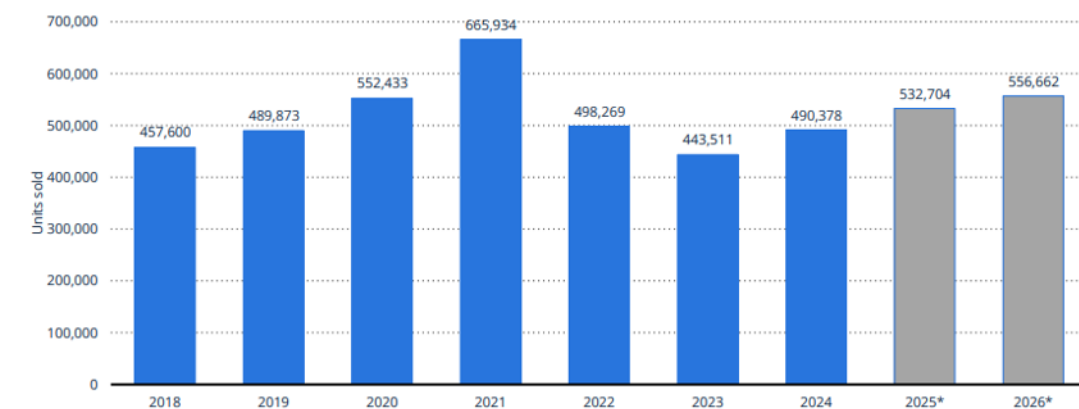
According to Statista, the residential housing market in Canada is facing steady growth since the peak in 2021 and the lowest rate in 2023. Several factors affect this market, such as annual income, house rental, interest rates, new residential houses construction and government intervention.

Analyzing the number of houses sold in Canada in the last nine years, we can note that 2021 was the year in which the most houses were sold with 664,934 units and the forecast for 2026 is to sell 556,662 units by until the end of the year. Comparing 2021 and 2026 information, there was a negative growth of 16%.

Figure 1 – Number of Housing Sold in Canada from 2018 to 2024

Number of housing units sold in Canada from 2018 to 2024, with a forecast by 2026

Number of house sales in Canada 2018-2024, with a forecast by 2026



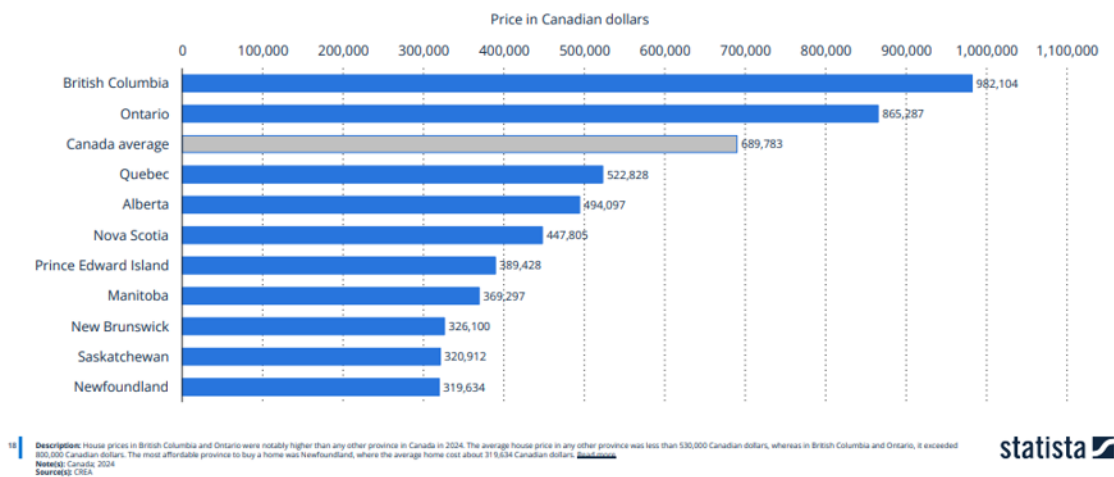
1 Description: After surging in 2021, sales activity in the Canadian housing market slowed down in the next two years. According to the forecast, the number of home sales in 2026 is expected to reach almost 556,662. The Canadian residential housing market is going through a period of change because the skyrocketing home prices are being tempered by various governmental interventions. One of the measures is such as a two-year ban on foreign purchases. Additionally, the government introduced a 5% GST on new homes.
 Notes: 2018 to 2024: * forecast as of January 2025
 Sources: CREA

According to a study, the average price of a house in Canada in 2024 was \$ 689,783, with British Columbia (\$982,104) and Ontario (\$852,287) having the highest average prices by province, followed by Quebec (\$522,828) and Alberta (\$494,097), respectively. On the other hand, Newfoundland (\$319,634) and Saskatchewan (\$320,912) had the lowest average house prices.

Figure 2 – Average House Prices in Canada in 2024

Average house prices in Canada in 2024, by province (in Canadian dollars)

Average house prices in Canada 2024, by province

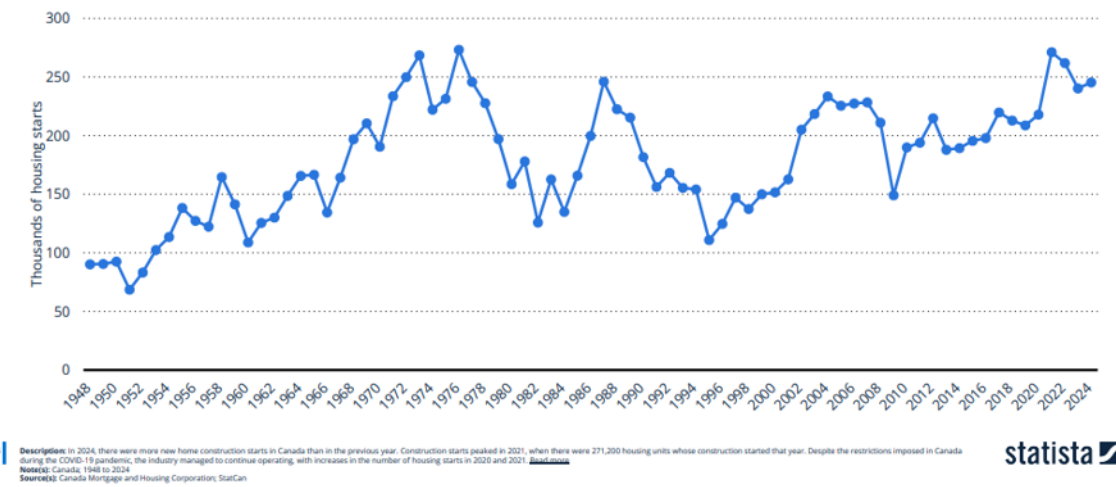


Regarding new residential construction in Canada, it is expected to grow after peaking in 2021 with 271,200 units, followed by a decline in 2022 and 2023, mainly due to the pandemic.

Figure 3 – New Housing Start in Canada from 1948 To 2024

New housing starts in Canada from 1948 to 2024 (in 1,000s)

New residential construction in Canada 1948-2024



The rental market is another factor that affects residential house prices, as when demand is high and supply is low, prices tend to increase.

Another important factor influencing housing prices and the market is the annual income needed to buy a house. Vancouver has the highest annual income (\$238,820), followed by Toronto (\$204,840). We can clearly see a direct link between annual income and average house prices across provinces and cities.

Research Gap

The literature shows that many articles use traditional statistical methods and machine learning techniques to analyze house prices in other countries, but there are a few studies on house price analysis using MLR or Random Forest in Canada. Research on Canadian housing price forecasting remains relatively limited compared with studies conducted in other countries (Brannlund et al., 2023). The Canadian case study differs from other countries because it has distinct characteristics, such as a high immigrant rate, regional price disparities, mainly in cities in British Columbia, Ontario and Quebec, and government regulations.

Some studies have used MLR to predict house prices, but some independent variables do not have linear relationships. In this case, the study's accuracy could be affected because non-linear relationships were not considered, such as location, market conditions, internal policies and others. Previous research has found that Random Forest often produces more accurate house price predictions than MLR. However, most of these studies have been conducted in house markets outside Canada (Zhong, 2024; Fu, 2024).

This analysis was performed to compare MLR and Random Forest predictions to determine the most effective approach for predicting house prices in Canada, analyzing house price variance according to location, characteristics and other variables.

Methodology

Data Selection

The dataset which was used in this paper includes a list of real estate properties in Canada, and was downloaded from Kaggle, a well-known open data platform for data science analysis. The dataset contains 44,896 records and 23 variables that describe physical, geographic and amenities of housing real estate across Canada's 10 provinces and territories, including British Columbia, Ontario, Alberta, Saskatchewan, among others. The relevance of this set relies on significant fluctuations in the price of Canadian real estate market during the last years. Price in housing considerably varies among regions which makes necessary to identify the factors that determine it to support buyers and investors for decision-making.

Tools & Techniques

To execute the analysis, Google Colab was chosen as the primary Python environment. For data handling the Pandas library was used, while Numpy was used for managing mathematical

computations. Additionally, Matplotlib and Seaborn was used for visualization. The core machine learning work was implemented via Scikit-learn. Statsmodels was used for statistical validation and Backward Elimination steps. (Harris et al., 2020; Matplotlib Development Team, 2025; pandas development team, 2024; Scikit-learn developers, 2025; Seabold & Perktold, 2010; Waskom, 2021).

Two predictive techniques for modeling were used. On one hand, Multiple Linear Regression was selected for its interpretability and capacity to model the relationship among multiple independent variables and the price of a property (James et al., 2023). On the other hand, Random Forest was selected for its ability to capture non-linear relationship and complex data patterns which a linear regression cannot detect. Both techniques were applied under the same clean dataset for further comparison between both approaches.

Data Analysis

Dataset Description

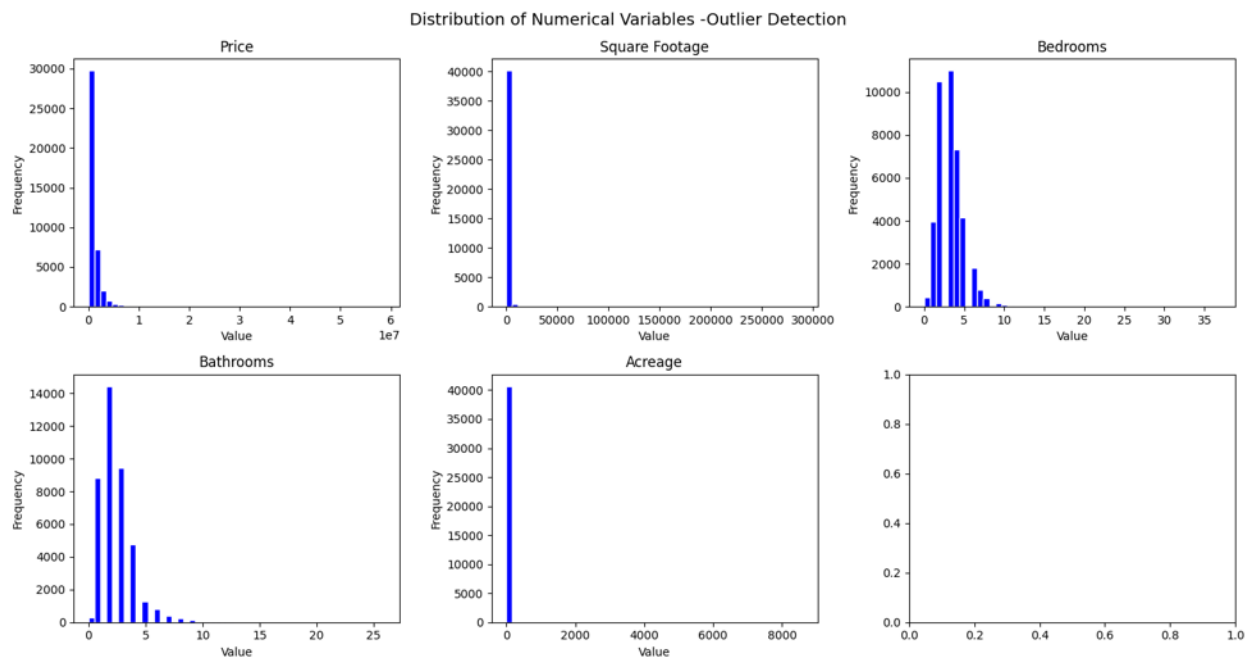
The dataset features 44,896 records of Canadian homes, each defined by 23 variables. We organized these into four main blocks. For location, we used variables like City and Province. The physical dimensions and price (Price, Acreage, Square Footage) form the numeric category. Additionally, we included seven binary indicators for amenities like Fireplaces or Gardens, along with multiclass variables such as Sewer and Property Type to capture more complex categorical details. As explained further, each of these variables had a different treatment during each method. The dependent variable of the model is Sales Price per residential property in Canadian dollars (CAD).

Data Cleaning

The following columns were eliminated because they had more than 50% of null values, which make them unreliable for the model: Roof (78%), Basement (67%), Flooring (66%), and Exterior (61%). The Heating variable showed 15% of null values, so by imputation with the mode, those values were filled with the most common value, forced air (Scikit-learn developers, 2025).

A total of 4,201 duplicate records were eliminated in order to reduce error in the model (pandas development team, 2024). The first appearance of each one of them remained in the model. For the **outliers**, the IQR method was applied to Price, Square Footage, Bedrooms, Bathrooms, and Acreage variables.

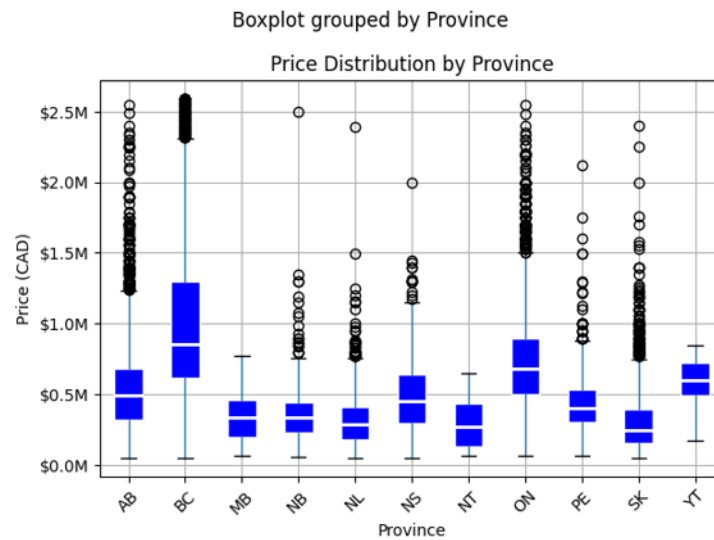
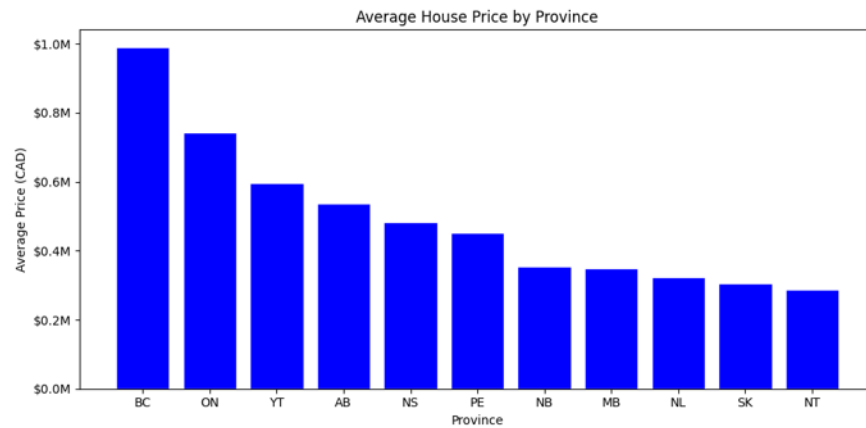
Figure 4– Distribution of Numerical Variables



The histograms confirmed a big positive skewness in all variables, which supports the decision of eliminating 9,723 records that presented outliers through the IQR method. Finally, after completing the data cleaning, the final dataset contained 30,972 observations and 19 columns, no missing values, no duplicate records, and no extreme outliers.

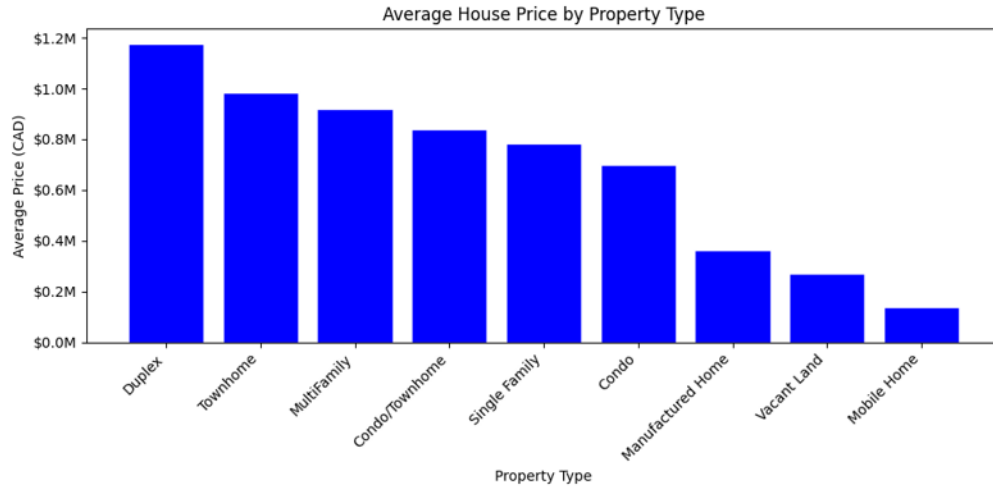
Exploratory Data Analysis

Figure 5- Price per Province



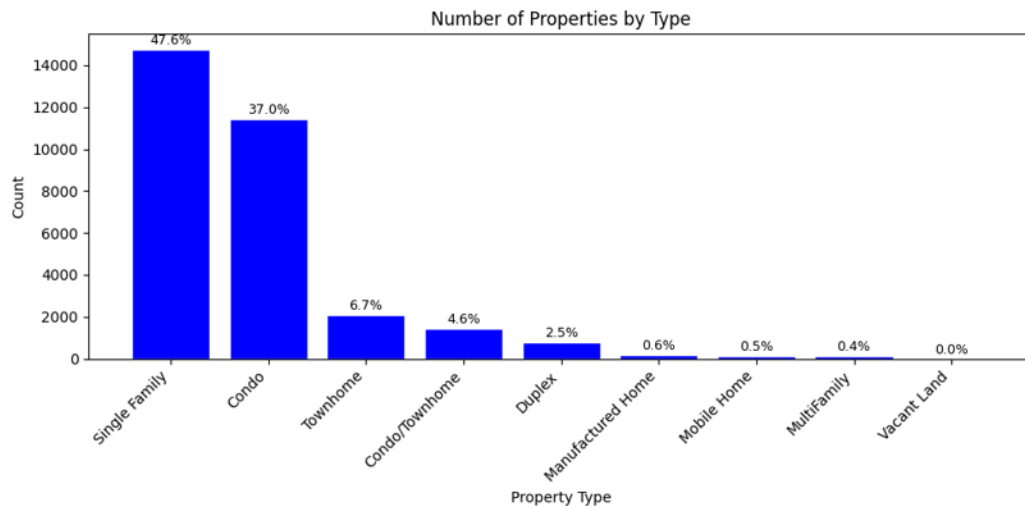
As expected, British Columbia and Ontario had the highest prices showing the influence of cities like Vancouver and Toronto. In contrast provinces like Northern Territories or Saskatchewan have lower prices suggesting big real differences in the Canadian real estate market. The boxplot reveals that British Columbia is the province with most variability and that has luxury residences that increases its distribution.

Figure 6- Price per Property Type



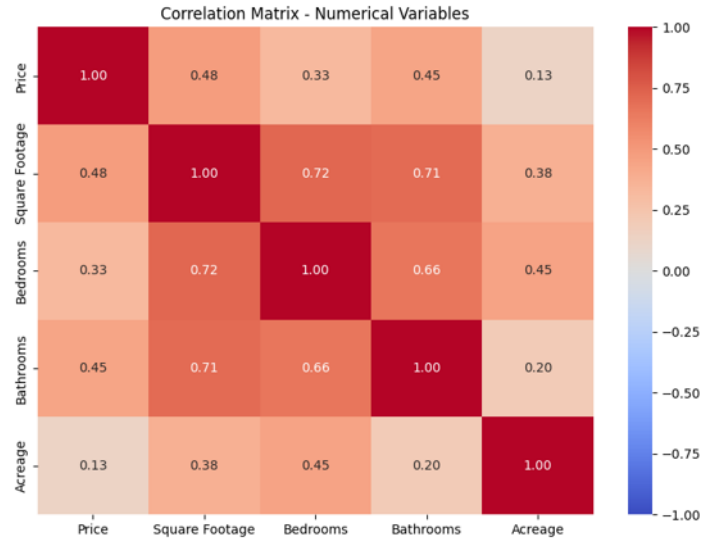
As shown in the graph, duplex and townhomes present the highest prices which can probably reflect their frequency in markets like Toronto or Vancouver. Vacant land and mobile homes are the cheapest, which could be consistent with their low demand in residential properties.

Figure 7- Property Type Distribution



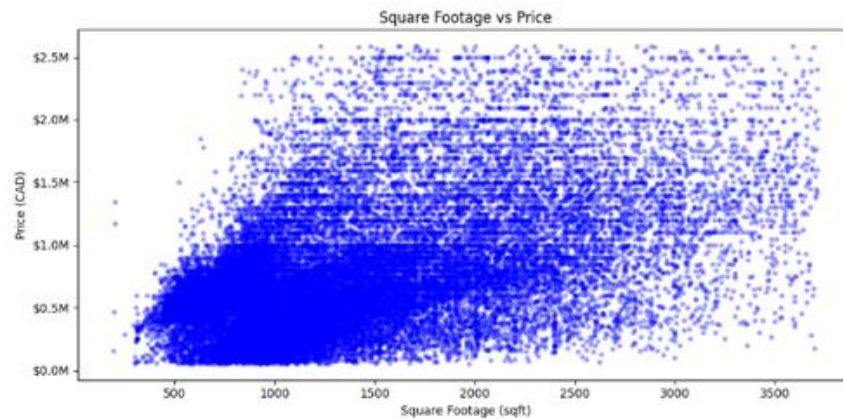
Single family and condos account for 84.6% of the total properties. The least common of the properties are vacant land, multifamily, and mobile home, which have few observations which might limit the ability of the model to predict prices for these types of properties.

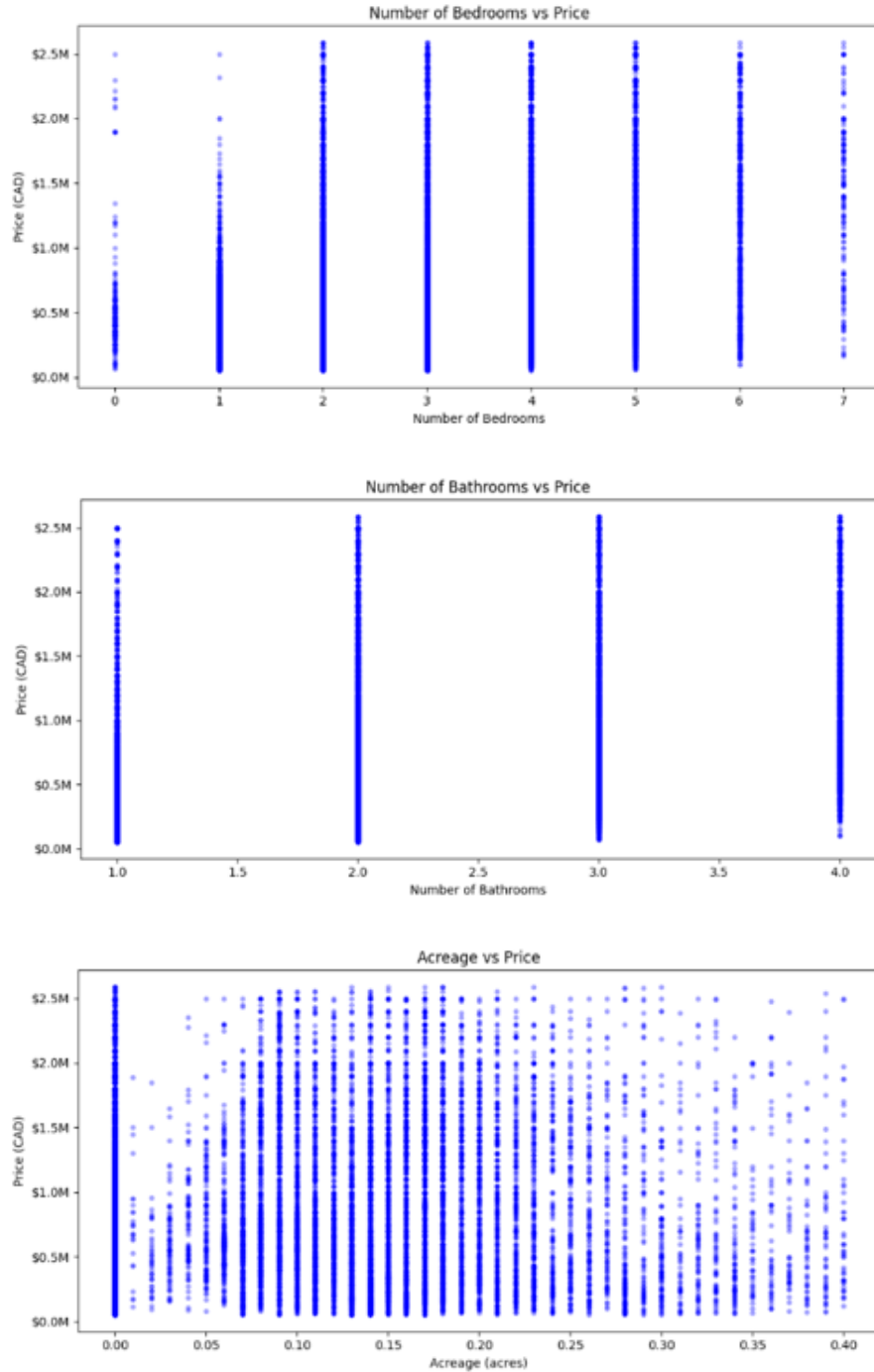
Figure 8- Correlation Matrix



Square Footage has the strongest positive correlation with Price (0.48) which is followed by Bathrooms (0.45), and Bedrooms (0.33). Nevertheless, it was observed some moderate correlations among Square Footage, Bedrooms, and Bathrooms (0.66-0.72) which suggests multicollinearity in the model and will be discussed during the Regression Model (Waskom, 2021).

Figure 9 – Relationship Among Numeric Variables and Price

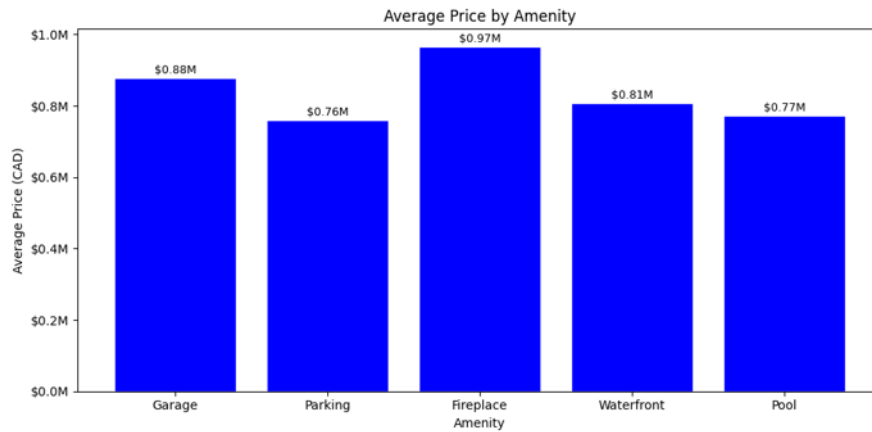




Square Footage vs. Price scatterplot shows a positive relationship among these two variables; however, a high dispersion is observed suggesting that this variable by itself does not completely explain the price. Bedroom, Bathrooms, and Acreage due to its discrete nature forms

vertical lines with moderate positive tendency which is consistent to their correlation coefficient.

Figure 10 – Average Price per Amenity



The bar chart clearly shows that properties that have fireplaces have the highest prices followed by garage and waterfront.

Multiple Linear Regression Model

For the Multiple Linear Regression model, binary categorical variables were changed to numeric (1/0) and the multiclass variables transformed into dummies by One Hot Encoding with `drop_first=True` to avoid having extra records in the dataset (Scikit-learn developers, 2025; pandas development team, 2024). We decided that City, Latitude and Longitude variables would be excluded from the model due to their high cardinality and non-linear nature. Besides, the dataset was divided in 80% training (24,777 observations) and 20% testing (6,195 observations). Furthermore, Backward Elimination was used to iteratively eliminate variables with p-values greater than 0.05 (James et al., 2023). Finally, 14 variables were excluded from the model including Pool, Fireplace, Garden and some categories of Heating and Sewer. The final model consisted of 27 variables.

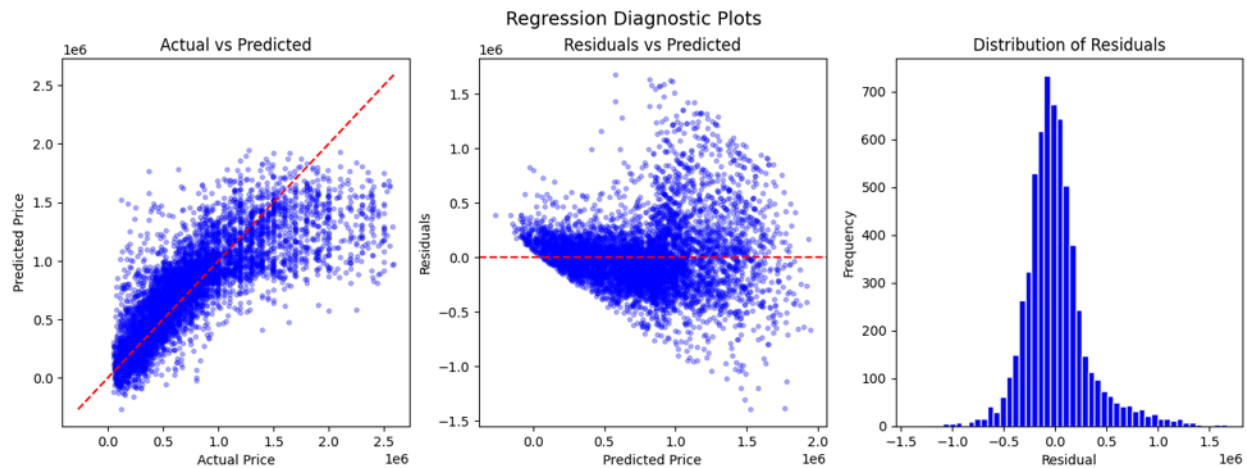
```

Final Model Summary:
                        OLS Regression Results
=====
Dep. Variable:          Price    R-squared:                0.610
Model:                  OLS      Adj. R-squared:             0.609
Method:                 Least Squares    F-statistic:          1433.
Date:                   Sat, 13 Jun 2026    Prob (F-statistic):    0.00
Time:                   04:06:05    Log-Likelihood:       -3.4882e+05
No. Observations:      24777    AIC:                  6.977e+05
Df Residuals:          24749    BIC:                  6.979e+05
Df Model:               27
Covariance Type:       nonrobust
=====
                        coef    std err          t      P>|t|      [0.025    0.975]
-----
const                -1.237e+05    9167.738     -13.493    0.000    -1.42e+05    -1.06e+05
Bedrooms              1.374e+04    2699.551       5.090    0.000     8450.531     1.9e+04
Bathrooms             8.259e+04    3624.865     22.784    0.000     7.55e+04     8.97e+04
Acreage               2.389e+05    3.37e+04       7.095    0.000     1.73e+05     3.05e+05
Square Footage        251.2525      5.246     47.892    0.000     240.970     261.535
Garage                1.055e+04    4986.639       2.115    0.034     772.679     2.03e+04
Parking              1.042e+04    4682.236       2.225    0.026     1239.095     1.96e+04
Waterfront            8.223e+04    3.63e+04       2.267    0.023     1.11e+04     1.53e+05
Balcony              8.143e+04    9488.042       8.582    0.000     6.28e+04     1e+05
Province_BC           5.772e+05    6574.140      87.805    0.000     5.64e+05     5.9e+05
Province_MB           1.684e+05    4.55e+04       3.698    0.000     7.92e+04     2.58e+05
Province_NB           7.582e+04    1.44e+04       5.275    0.000     4.76e+04     1.04e+05
Province_NS           1.606e+05    1.34e+04      12.003    0.000     1.34e+05     1.87e+05
Province_ON           4.757e+05    9273.234     51.303    0.000     4.58e+05     4.94e+05
Province_PE           1.133e+05    1.8e+04       6.277    0.000     7.79e+04     1.49e+05
Province_SK           -1.712e+05    8745.261     -19.581    0.000     -1.88e+05     -1.54e+05
Property Type_Condo/Townhome -6.511e+04    1.03e+04     -6.291    0.000     -8.54e+04     -4.48e+04
Property Type_Duplex   2.276e+05    1.42e+04     16.078    0.000         2e+05     2.55e+05
Property Type_Manufactured Home -4.571e+05    2.56e+04     -17.836    0.000     -5.07e+05     -4.07e+05
Property Type_Mobile Home -1.05e+05    3.02e+04     -3.476    0.001     -1.64e+05     -4.58e+04
Property Type_Single Family 4.446e+04    7751.303       5.735    0.000     2.93e+04     5.96e+04
Property Type_Townhome -6.829e+04    9080.853     -7.520    0.000     -8.61e+04     -5.05e+04
Heating_fireplace     -1.641e+05    4.77e+04     -3.439    0.001     -2.58e+05     -7.06e+04
Heating_heat pump     1.038e+05    7674.225     13.531    0.000     8.88e+04     1.19e+05
Heating_other         3.71e+05    5.86e+04       6.330    0.000     2.56e+05     4.86e+05
Heating_radiant       -6.076e+04    5437.238     -11.175    0.000     -7.14e+04     -5.01e+04
Sewer_municipal       -2.825e+05    6110.671     -46.233    0.000     -2.94e+05     -2.71e+05
Sewer_septic          -2.413e+05    1.22e+04     -19.824    0.000     -2.65e+05     -2.17e+05
=====
Omnibus:              4911.130    Durbin-Watson:         2.000
Prob(Omnibus):        0.000    Jarque-Bera (JB):      15227.892
Skew:                 1.020    Prob(JB):              0.00
Kurtosis:             6.253    Cond. No.              4.66e+04
=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 4.66e+04. This might indicate that there are
strong multicollinearity or other numerical problems.

```

Regression Diagnostic Plots

Figure 11 – Regression Diagnostic**Figure 12 – Predicted Housing Prices**

First, the Actual vs. Predicted Price scatterplot shows a clear linear relationship confirming the linearity assumption; nevertheless, it was observed greater dispersion among properties with high prices suggesting that the model predicts best low to medium prices. Second, the Residual vs. Predicted scatterplot reveals a funnel pattern, suggesting heteroscedasticity since the spread of residuals increases for higher predicted values (Statistics By Jim, 2025). This indicates that prediction errors are larger for expensive properties. Third, the histogram of residuals shows an approximate normal distribution with some positive asymmetry, partially covering the normality assumption (Hunter, 2007).

Moreover, the Actual vs. Predicted Housing Prices for a sample of 100 properties reveals that the model closely follows the general price tendency. Both the blue and the red lines are close to each other in properties with middle prices; however, the greatest deviations occur in extreme prices. This confirms that the model is more reliable on predicting properties in the regular price range of the Canadian market (Harris et al., 2020).

Multicollinearity

Because of the high correlation among Square Footage, Bedrooms, and Bathrooms, a VIF was run for the three variables showing values greater than 10 (Hair, 2019). It was evaluated the elimination of Bedrooms from the model, since it was the least correlated to price, but VIF did not improve considerably and R² was reduced by 0.0004. It was decided to keep the variable and acknowledge multicollinearity as a limitation of the model due to the nature of the real estate industry.

Model Evaluation

```
FINAL MODEL EVALUATION
R2:                0.6053
Adjusted R2:       0.6036
RMSE:              $312,921.74
MAE:               $217,642.33
Average Price:     $770,092.56
RMSE as % of avg:  40.6%
F-statistic:       1433.09
p-value (F):       0.000000
Variables in model:27
```

The final model explains 60.5% of the variation in the price of real estate properties across Canada. Significant F p-value of 0.0000 confirms that the model is statistically significant, and the minimal difference between R² and Adjusted R² confirms that the model is not overfitting. RMSE of \$312,921.74 is 40.6% of the average price which shows the complexity of the Canadian

real estate market and the fact that there are factors not included in this model that affect the price of the residential units.

Random Forest Model Analysis

Residential housing prices were predicted throughout Canada using the Random Forest model. The first attempt was based on LSTM neural network, but this type of network was not suitable for the collected dataset as there was no sequential or time dependent structure. Random Forests are, on the other hand, capable of handling tabular, heterogeneous data and can handle nonlinear relationships without need of any temporal dependences.

Data Preprocessing

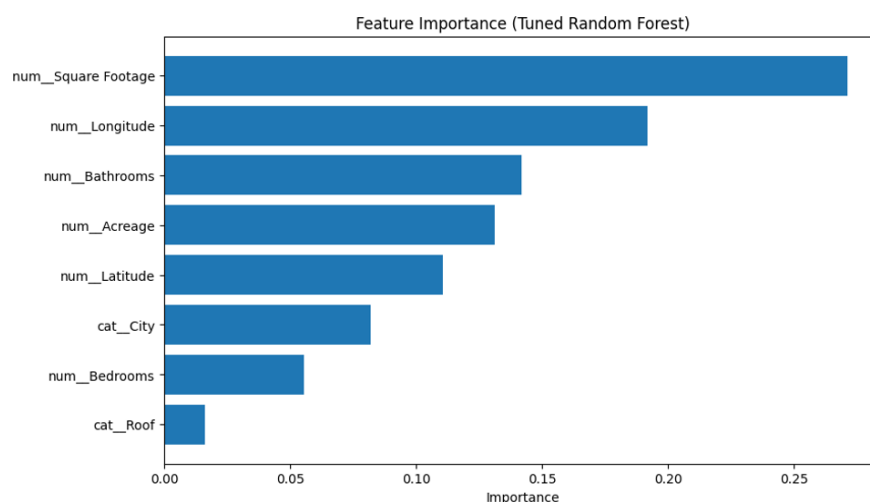
The preprocessing stage began with the cleaned dataset produced earlier in the project. All the variables were partitioned into numerical and categorical variables to handle them correctly while training the model. Categorical variables were encoded with an Ordinal Encoder which was efficient to compute and suitable for models based on trees because categories in the variable were not forced to be linearly related. Following this, the dataset was split into 80% training and 20% testing to facilitate the proper evaluation of the model performance. Furthermore, it was found that schema consistency between the inputs into the training and prediction phases was important for ensemble models that necessitate the same feature structure.

Feature Engineering

To make the model easier to interpret and more accurate to predict, feature engineering was performed. The categorical variables were kept as they were high cardinality variables and Random Forest can easily deal with large categorical spaces. Geographic indicators (Latitude, Longitude) were added in to account for geographic variation, while binary amenities (e.g., Garage, Fireplace, Parking) were retained as indicator variables. A baseline model was trained

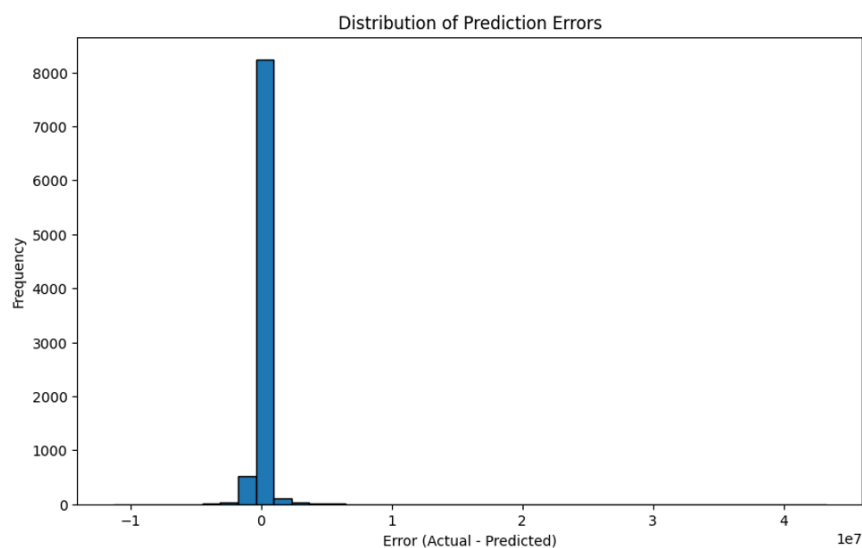
first, feature importance scores were extracted to identify variables contributing less than 1% to predictive performance. The low impact features (e.g., Roof, Exterior, Garden, Pool, and Sewer) have been removed to eliminate noise and speed up computational speed.

Figure 13 – Feature Importance



Note. The chart shows the strongest variables, supporting the decision to remove low-importance variables.

Figure 14 – Distribution of Prediction Errors



Note. The residuals are centered around zero, indicating unbiased predictions, while the long right tail reflects the model's difficulty predicting high-priced homes.

Model Development and Hyperparameter Tuning

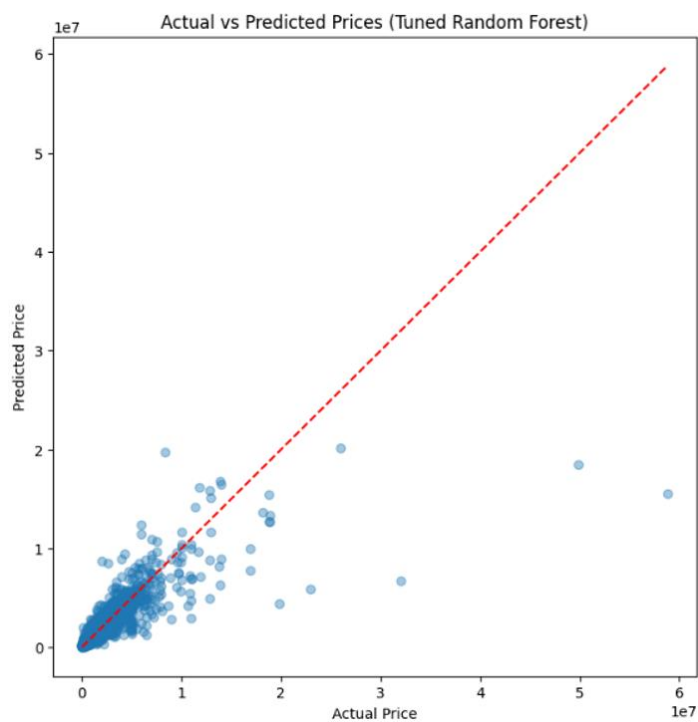
The baseline Random Forest model was trained using 100 trees, a maximum depth of 20, and minimum split and leaf constraints to prevent overfitting. This first model yielded an RMSE of 883,691, MAE of 223,999, and R^2 of 0.6989 as a benchmark to optimize further. Features with low importance were removed and a tuned model was created with 400 trees, a maximum depth of 25 and the “sqrt” strategy during a split. This led to increased model stability and prediction accuracy, with an RMSE of 855,394, an MAE of 217,154, and an R^2 of 0.7179. The increase in R^2 shows that the tuned model accounts for a higher percentage of the variance in housing prices, and the decrease in RMSE and MAE shows that the model's predictions are more accurate across the entire dataset.

Table 2 – Comparison of Baseline and Tuned Random Forest Models

Model	RMSE	MAE	R^2
Baseline Random Forest	883,691.79	223,999.36	0.6989
Tuned RF (Reduced Features)	855,393.98	217,154.27	0.7179

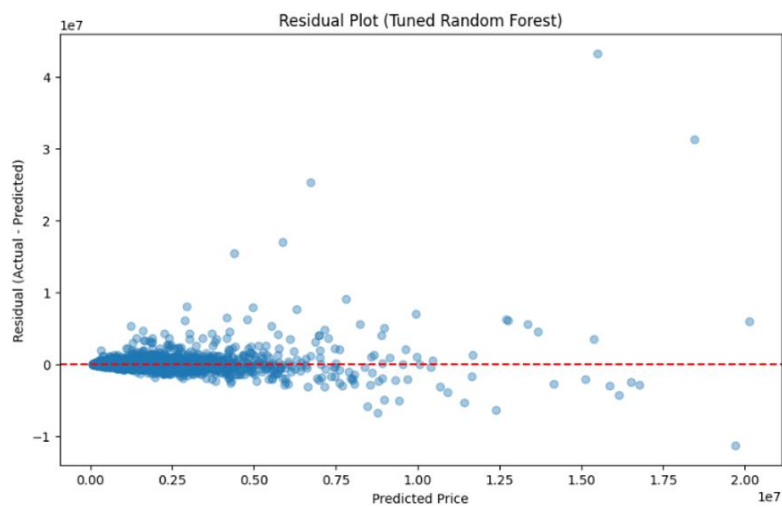
Note. RMSE = Root Mean Squared Error; MAE = Mean Absolute Error; R^2 = Coefficient of Determination.

Figure 15 – Predicted Prices



Note. Points close to the diagonal line indicate accurate predictions, while the spread and visible clusters reflect model error and underlying market segments.

Figure 16 – Residual Plot

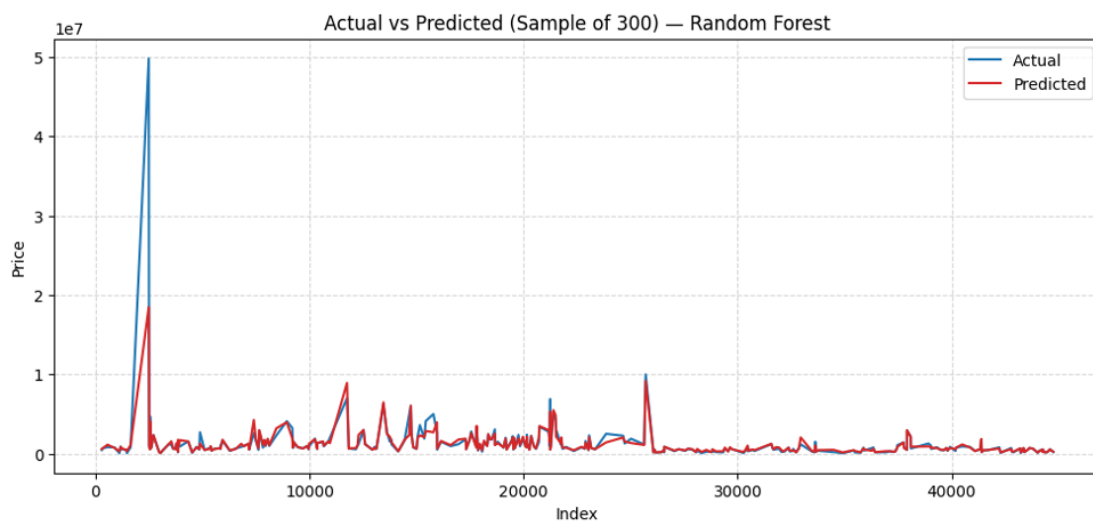


Note. Random scatter indicates a well-behaved model, while visible patterns or a funnel shape suggests missing structure and heteroscedasticity.

Model Performance Evaluation

To visualize the model's performance, a chart was created comparing actual and predicted prices for a sample of properties. This visualization shows how close the model is to general pricing trend and larger deviations for high value properties.

Figure 17 – Predicted Prices



Note. This chart shows continuous actual and predicted price lines with a clear train–test split, illustrating how closely the Random Forest model tracks real values despite not being a time-series model.

The model proved to be very successful for homes in the low to mid-range price segment, consistent with the distribution of the Canadian housing market. The greater the deviation in luxury properties, the more natural and complicated real estate prices can be.

Comparison with Regression

The performance of the Random Forest model was better than the Multiple Linear Regression model. The regression model obtained an R^2 of 0.605 and an RMSE of \$312,921, when the cleaned data was used. While the RMSE values cannot directly be compared (because of the

differences in the size of the data sets and how the outliers were treated), the higher value of the R^2 obtained by the Random Forest suggests a better capacity of the model to capture the nonlinear relationships and complex interactions between variables. Moreover, unlike regression model which were shown to have restrictions on linearity, normality and multicollinearity (Hair et al., 2019), the Random Forests model does not assume any of these.

The findings from the Random Forest also aligned with economic intuition, with Square Footage, Latitude, Longitude, and Acreage being the most important factors to predict price. Overall, the Random Forest model gave a flexible and data driven model that worked well with the interpretability of the regression model. It proved to be a useful tool in the analytical framework of the project as it could process high dimensional feature space and nonlinear patterns.

Conclusion and Recommendations

This study explores a number of factors that affect the residential housing prices in Canada. A predictive model has been developed and compared to the traditional Multiple Linear Regression model in order to better evaluate several different approaches for the estimation of the residential housing prices in Canada. By utilizing a database of approximately 45,000 residential listings across all of the Canadian provinces and territories, this study includes an exploratory data analysis of the key factors affecting the residential housing prices in Canada, a Multiple Linear Regression model that includes a series of key factors affecting the residential housing prices in Canada, and an evaluation of several predictive models including Random Forest and an LSTM neural network in order to better identify the most appropriate model for estimating the residential housing prices in each of the individual markets across Canada.

The results have provided conclusive evidence regarding the complex interplay of property attributes, location, and regional real estate market(s) as determinants of residential property values

across Canada. The strongest factor influencing the price of a residential property has been determined to be the Square Footage, or size of the property, which was identified in every stage of analysis. Other influential factors include indicators of location including the province in which the property is located, latitude and longitude, and acreage. The exploratory data analysis results provide further support to earlier findings in terms of the significant variations in between the high value real estate markets in the Provinces of British Columbia and Ontario, and the lower valued real estate markets found in the majority of the other provinces across the country. The findings illustrate that in addition to Square Footage, all residential properties are also influenced by a number of other key factors. The combination of these factors can create significant differences in between the values of individual properties, which cannot be explained by means of a single variable. As a result, a number of property characteristics, as well as several key location indicators, are used together to evaluate and assess the value of individual residential properties found throughout the entire of Canada.

Through Multiple Linear Regression the strongest predictors of property value were established and a model created that explained just over 60.5% of the variance in the values of residential properties in Canada. This study found that as a result of the high number of interacting variables many of the key variables were highly inter-correlated leading to issues with multicollinearity and as a result decreased the reliability of the estimates for some of the key variables. The study also found that the residuals from the model were not evenly distributed leading to the existence of heteroscedasticity. Overall while Multiple Linear Regression provided a reasonable model of residential property values across Canada it did not capture enough of the variation to be of use in a number of specific situations. In particular the model performed very poorly in a number of instances where very high value residential properties were involved. The

study found that in a number of instances the model failed to even come close to estimating the value of such properties and as a result it appears that residential properties in Canada can have values that cannot be predicted through the use of a linear model.

The Random Forest model performed better than the other models in terms of predictive accuracy for this data set. It was particularly good at modeling complex patterns in the data without requiring the analyst to make strong assumptions about the relationships. In particular, the model was able to model non-linear relationships very well, and it was able to take into account interactions between the various input variables. The model's performance is consistent with the results obtained for the other models in terms of the variables that were most important for modeling the housing values. However, the Random Forest model performed better than the other models in terms of identifying the location and property size indicators as the most influential variables for modeling the housing values. Thus, this model is more suitable than an LSTM neural network for this data set, since the data set does not contain a large number of time steps or a time-dependent structure. This study highlights an important lesson in analytics: the most complex model is not always the best model. The choice of a suitable model is highly dependent on the characteristics of the data as well as the objectives of the analyst. In general, the choice of a model should be based on the following criteria: simplicity, interpretability, and adequacy.

In addition to providing conclusions with respect to comparison of the models, the results of this research also illustrate the evolution of the approach to property valuation in real estate decision making. Historically, valuation of properties has relied on the appraisal methods such as using comparable sales as well as on the expert's judgment. However, given the level of complexity that has been introduced into the Canadian real estate market in recent years, there are relationships between variables that cannot be identified through conventional analysis. Thus, the

application of predictive analytics should not be confined to enhancing existing analysis. Rather, it should be viewed as a strategic capability that supports more objective, informed and efficient decision making in the real estate market.

Accuracy of prediction and interpretability of results are not mutually exclusive; both can be important in a number of different contexts. The Multiple Linear Regression model in this study, for example, is transparent and provides clear results that can be used for financial reporting, for example, and for policy evaluation. On the Random Forest model, on the other hand, provides strong results in terms of forecasting accuracy. Thus, the choice of model will depend on the analyst's goals in developing a model.

The use of machine learning models for prediction in residential housing markets, such as the one studied here, can be a valuable addition to current models and forecasting methods. Many current methods for residential housing market prediction are based on historical data, often utilizing linear relationships that fail to capture much of the variation in the data. Studies that utilize machine learning for residential housing markets have reported superior performance to traditional methods, such as Multiple Linear Regression. The study reported here uses two different approaches to prediction in the residential housing market of Canada and reports on the strengths and limitations of each. Results indicate that the Random Forest model is superior for predicting residential housing prices, due in large part to the model's ability to capture non-linear relationships in large datasets. Importantly, the model performed well with respect to a number of indicators of model performance and was able to provide interpretable results regarding the relationship between the input variables and the predicted variable. These results are particularly important, given the lack of recent studies that have utilized machine learning for residential housing market prediction in Canada. In summary, this study is an important contribution to the

growing body of housing market analytics research in Canada, and suggests that the use of machine learning models, such as the Random Forest model, can be particularly valuable in complex residential housing markets.

While this study aimed to compare two possible models for housing price estimation, the most important finding for anyone involved in Canadian real estate is that our housing market is becoming more complex and will need to be analyzed with tools greater than simple valuation. Each property has many indicators and in the end it is a complex of location indicators and property size indicators (and possibly other indicators that were not included in this study) that can be modeled in order to create the best estimate of the price of a house. In this way, such a model can transform all of the housing data available to a company or government into solid business intelligence, that can be used for a variety of purposes, such as making better investment decisions, assessing greater risk in lending, identifying greater areas of affordability problems, supporting better housing policy, and informing the best possible use of funds in support of affordable and special needs housing.

Managerial Recommendations

The findings of this study have lead to several conclusions for all stakeholders involved in the Canadian real estate market.

1. Integrate predictive analytics into property valuation processes.

Traditional property valuation approaches can be supplemented and improved by using predictive analytics tools that process large amounts of property and market data to produce more accurate property valuations.

2. Strengthen lending and risk assessment practices through advanced modelling.

For financial institutions, such as banks and other mortgage lenders, advanced modeling could be used in conjunction with their current property valuation processes to better support lending decisions. By identifying the true value of a property and incorporating additional credit risk factors, such models can assist in the process of evaluating loan applications, improve lending decisions, and also help to manage risk and reduce exposure to fluctuations in the housing market.

3. Adopt regionalized investment strategies.

For Investors: Support localized investment decisions by helping to understand regional and forecast the performance of properties within different markets and geographic locations. For instance, data about the different provinces and major metropolitan regions in Canada reveals that certain geographic variables are some of the strongest determinants of housing prices within the Canadian real estate market. Therefore, investors should not rely solely on the performance of the national housing market. Rather, investors should take into consideration local trends that can impact housing prices in different regions across the country.

4. Utilize predictive analytics to support housing policy development.

Predictive models can also be used by Government agencies and housing authorities to determine and forecast affordability issues in different markets. They can also use these models to forecast future demand for housing in various markets. They can also use these models to forecast the impacts that policy interventions could have in various housing markets.

5. Establish model governance and continuous monitoring processes.

Establish a model governance framework to validate, monitor and retrain models periodically, as housing markets evolve with changes in the economy, demographics and government regulations.

Limitations

All research has limitations and those for this research are as follows.

First, property listings typically consist of advertised or advertised prices, not transaction prices. In addition, property listings are usually residential properties while business or land properties are often included as well.

Second, the analysis focused primarily on the properties' characteristics and the geographic location where they are found. As mentioned earlier, there are many other macroeconomic factors that can affect the prices of houses, such as the interest rates, inflation, employment conditions, the mortgage affordability, the immigration trends, the household's income, and the population growth, that were not included in the models.

Third, the models only incorporated a set of property-level attributes and a set of regional attributes, while a number of additional factors—such as interest rates, inflation, employment, immigration, household income, and population growth—could likely have a very large impact on the prices of individual homes. The exclusion of these additional factors from the models presented here likely has a substantial impact on the results, especially given that the models are only fitted to a cross-sectional dataset rather than to a series of time-series observations on individual homes.

Fourth, all the data used for this study was in the form of cross-sectional data. The models developed in this study will only explain the price of the houses in the study for the time period the data was collected for. The study will not be able to explain how the housing market behaves over time.

Future Research

Further, future research could utilize a combination of cross-sectional and time-series data (i.e. a longitudinal study) in order to forecast the future values of houses as well as examine how various factors could affect housing prices over time, such as interest rates, economic conditions, and even policy decisions.

Other characteristics that would be of interest to include in a housing market prediction model are macroeconomic variables, as well as demographic indicators. Some possible indicators to consider would be household income, employment rates, inflation, as well as population growth and change. Additionally, data could be included to track immigration to different parts of the country and how this may affect local markets. Housing prices are also affected by the interest rates on mortgages, and this could also be an interesting indicator to include.

The authors also suggest for future research the comparison between additional models, such as the Gradient Boosting, XGBoost, and the LightGBM models, and other ensemble models that are currently performing well in data that is comprised of many features and that are of a structured nature. Such future research would identify the best model for use in forecasting the behavior of the Canadian housing market.

Finally, future studies could also be developed on a province by province basis as well as on a metropolitan basis. Given the extreme regional variation found within this study, it would be very useful to develop more accurate models that can be used by investors, lenders, real estate brokers and agents, as well as by government and non-government housing policy makers to forecast future prices in the various markets across Canada.

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